Module 03: Perception in Robotics — Autonomous Sensing

Learning Objectives

1. Define **autonomous perception** as the capacity of a robot to build and maintain a comprehensive world model from its own sensor data without human labeling or supervision.
2. Analyze how **SLAM, object detection, and semantic segmentation** implement the perception component of autonomous Active Inference.
3. Apply the concept of **perceptual robustness** to understand failure modes of autonomous perception systems.

Introduction

An autonomous robot perceives the world not through a single sensor or a single algorithm, but through a **multi-modal, multi-scale perceptual system** that constructs and maintains a rich internal model of the environment. This model must support navigation (where am I, where can I go?), interaction (what objects are here, what can I do with them?), and prediction (what will happen next?).

Key Concepts

1. SLAM: Simultaneous Localization and Mapping

SLAM is the canonical perception problem of autonomous robotics — building a map of an unknown environment while simultaneously tracking the robot's position within that map:

- **Visual SLAM**: Uses camera images to extract features (ORB-SLAM, LSD-SLAM), track them across frames, and triangulate 3D structure
- **LiDAR SLAM**: Uses laser scans to match geometric features and build 3D point cloud maps

- **Graph SLAM**: Represents the pose history as a graph and optimizes the entire trajectory when loop closures are detected

In Active Inference terms, SLAM is joint inference over the hidden state (robot pose) and model parameters (map) — simultaneously estimating where you are and what the world looks like, using the consistency between observations across time as the binding constraint.

2. Object Detection and Recognition

Autonomous robots must identify and classify objects in their environment:

- **Detection**: Where in the sensor data is there an object? (bounding boxes, segmentation masks)
- **Recognition**: What kind of object is it? (car, person, chair, obstacle)
- **Pose estimation**: Where is the object in 3D space, and what is its orientation?

Modern deep learning approaches (YOLO, Mask R-CNN, PointNet) implement high-dimensional likelihood models — the A matrix mapping from hidden object properties (type, position, orientation) to pixel patterns or point cloud features, learned from large datasets.

3. Semantic Mapping

Semantic mapping extends SLAM by labeling the geometric map with semantic categories:

- The map contains not just "there is a surface at coordinates (x, y, z)" but "this surface is a sidewalk (traversable), this is a wall (non-traversable), this is a door (openable)"
- Semantic labels provide **affordance information** directly — the map tells the planner what actions are available at each location
- Active Inference interpretation: semantic mapping enriches the generative model with categorical hidden states that support richer policy evaluation

4. Perceptual Failure Modes

Autonomous perception systems fail in predictable ways:

- **Adversarial conditions**: Sun glare, fog, heavy rain, snow degrade visual and LiDAR perception — precision drops, uncertainty grows

- **Out-of-distribution objects**: The perception model was trained on specific object categories; novel objects (an overturned shopping cart, a large bird, a construction sign) may be misclassified or missed entirely
- **Sensor degradation**: Lens fouling, mechanical misalignment, electromagnetic interference reduce sensor quality
- **Perceptual aliasing**: Two different locations look identical to the sensors (identical corridors, repeated architectural features), causing the agent to confuse its position

Robust autonomous systems detect perceptual failure through **meta-perceptual monitoring** — tracking the confidence of their own perceptual estimates and triggering conservative behavior when confidence drops.

Applications

- **Last-mile delivery robot**: A sidewalk delivery robot uses visual-inertial SLAM for localization, semantic segmentation to distinguish sidewalk from street from grass, and pedestrian detection to avoid collisions — all integrated into a single perception pipeline that feeds the navigation planner.
- **Underwater inspection**: An autonomous underwater vehicle inspecting bridge pilings uses sonar SLAM (visual perception is degraded by turbidity) combined with learned defect detection — identifying cracks, corrosion, and biofouling from sonar returns.

Conclusion

Autonomous perception — SLAM, object recognition, semantic mapping, and robustness monitoring — provides the foundation for self-governing robotic action. The perceptual system constructs and maintains the generative model that all other components (cognition, action, planning) depend on. The next module examines autonomous cognition.